New IND-CPA-D Attacks on BFV and BGV

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A Artifact Appendix

A.1 Abstract

The artifact is a single Docker image that carries the three different attacks of the paper, each mounted on the homomorphic-encryption library it targets: the BGV bootstrapping attack of Section 3 on HElib, and the two BFV attacks of Sections 4 and 5 on (two different versions of) OpenFHE. The image builds all three libraries from pinned upstream commits and overlays only the attack code.

Running it produces the logs behind the paper's figures and tables, draws those figures, and writes a short report that sets each result beside the paper's.

This version of the appendix describes commit `03a1560`.

A.2 Description & Requirements

The artifact is the public repository on GitHub, and needs no data to run: the bootstrap-failing ciphertexts of Section 3, for its two parameter sets, are gathered by the default `make c1`, in about two hours. Optionally, the samples we gathered, behind the counts of Table 2, are available separately at OSF as `bgv-samples-m21845.tar` and `bgv-samples-m28679.tar`, 2 GB each.

The repository is structured as: i) `artifact/` holding `Dockerfile`, `Makefile`, `README.md`, and three short scripts that convert the samples, save figures and write the report. ii) `BGV/` holds the Section 3 attack, as sources overlaid on HElib, with the analysis of the failures; iii) `BFV-pt-ct/` and `BFV-ct-ct/` hold those of Sections 4 and 5, each with the script that turns a run's log into the paper's table or figure.

Both BFV directories carry a `paper_results/` of the logs behind the paper's figures and tables, while a fresh run writes to `results/`, so the two can be set side by side; for Section 3 the equivalent is the optional samples. The appendix experiments, which belong only to the full version, are included as well.

A.2.1 Security, Privacy, and Ethical Concerns. The attacks run entirely against keys and ciphertexts the artifact generates for itself, and nothing reaches the network after the image is built. No security mechanism on the evaluator's machine is touched or disabled.

The attacks run at specific parameters, and both libraries have been notified; the fixes are discussed in the paper.

A.2.2 How to Access. The primary, public repository for this artifact is hosted on GitHub at https://github.com/hmchoe0528/NEW_INDCPAD_HE, which is the address cited in the paper and where the code will be maintained.

Due to the coordinated disclosure process, the artifact was kept private until 19 October 2026. Therefore, the artifact evaluation was conducted using an Open Science Framework (OSF) project, which contains a copy of the code as well as the optional pre-collected samples. At the time of writing, this project is publicly accessible at https://osf.io/amhr5/. It is not a permanent archive, and we did not ask for the Artifacts Available badge.

A.2.3 Hardware Dependencies. We consider a standard x86-64 or arm64 machine, typically a regular laptop. Everything is compiled from source inside the image.

10 GB of RAM covers everything, and 4 covers all but Section 3, whose analysis memory-maps its samples rather than reading them. The highest memory we measured was 7.2 GB.

On disk, on top of the image, a fresh `make c1` writes 12 GB, and the samples take 4 GB.

Note that on Docker Desktop, the virtual machine carries its own limit (see Settings, Resources to adjust).

A.2.4 Software Dependencies. We require Docker, and nothing else. The two OpenFHE versions, commits `2bf4524` (v1.3) and `df495ba` (v1.5.0), matter: each binary is linked against its own.

A.2.5 Benchmarks. Section 3 needs the collected ciphertexts that failed to bootstrap. Gathering by oneself is what `make c1` does: a failing ciphertext costs $\approx$ 2.4 seconds, on top of $\approx$ 30 minutes to setup the library and the parameters. `AUTHORDATA=1`, and `make quick` if one is there, instead read a stored set: one gathered earlier by `make c1`, or our optional samples.

Sections 4 and 5 need no data. Each run generates its own keys and ciphertexts, so every number the report gives is computed by the run that wrote it.

A.3 Setup

A.3.1 Installation. Clone the public repository, and build the image. The build takes several minutes on an Apple M4 Pro:

```
cd NEW_INDCPAD_HE
docker build -f artifact/Dockerfile \
             -t indcpad-artifact .
```

Optionally, put the samples in `data-collected-by-authors/`.

Each experiment below is the same command with `TARGET` replaced by the target it names. The results directory is always mounted; the samples are mounted only for `make quick` and `AUTHORDATA=1`:

```
docker run --rm \
  -v "$PWD/results:/artifact/results" \
  -v "$PWD/data-collected-by-authors:\
      /artifact/data-collected-by-authors:ro" \
  indcpad-artifact make TARGET THREADS=8
```

A.3.2 Basic Test. `make help` prints all targets. `make quick` is the smoke test: it runs the attacks in quick mode in a few minutes and writes `results/REPORT.md`. Section 3 runs at the paper's parameters and gives Table 2, on a stored set of samples, and is skipped without one; the Section 4 part ends with a match ratio of `1.000000`,

drawing another key when the first is one the parameters do not cover; and Section 5 gives a count of separating key pairs. It validates no claim on its own, but shows that the image works.

A.4 Evaluation Workflow

A.4.1 Major Claims.

(C1): An IND-CPA-D adversary puts all 32 nonzero positions of the BGV secret key among 97 candidates, from the intermediate values of failing bootstrappings, for the parameters given in Section 3, as reported in Table 2 and Figure 2, based on an experiment (E1). The step that turns the candidates into the key, described in Section 3, is not run.

(C2): An IND-CPA-D adversary recovers the whole BFV secret key through plaintext-ciphertext (PT-CT) multiplication, within a depth the noise bound certifies, in N decryption queries. This is for Section 4, reported in Table 3 and Figure 3, based on an experiment (E2).

(C3): The same holds for ciphertext-ciphertext multiplication: additions seed a message-dependent noise and repeated squarings amplify it past the correct decryption threshold while the certified bound stays inside, which breaks indistinguishability. This is for Section 5, reported in Figures 4 and 5, based on an experiment (E3).

A.4.2 Experiments. Every target writes its logs and figures to `results/` and appends to `results/REPORT.md`, which sets each result beside the value the paper reports. The `results/REPORT.md` is what to read; it may navigate to logs and figures.

Passing `THREADS=8` is recommended. The timings are **not the claim**: what matters is that the attacks succeed at all.

We have three main courses to run, as shown below. The third covers the appendices, which the camera-ready does not carry.

course	command	time
basic test	`make quick`	few min
camera-ready	`make c1 c2 c3`	5.5 h
full version	`make appendix`	1.7 h

(E1): *BGV bootstrapping: Table 2 and Figure 2* (C1) [2.3 compute-hours (20 minutes with `AUTHORDATA=1`)]

Execution: `make c1 THREADS=8`; it gathers its own failing ciphertexts. Adding `AUTHORDATA=1` and the samples mount reads a stored set instead.

Results: The report gives Table 2, where the number of correct guesses climbs as the ciphertexts accumulate, from 50 to 500 in steps of 50; on a fresh gathering, expect 30 to 32 of 32 at Table 2's counts. With `AUTHORDATA=1` on our optional samples, it instead gives the same counts as Table 2. Figure 2 is the four panels `results/figures/fig2-0*.pdf`.

Remarks: Table 1 of the camera-ready gives $(h, \tilde{h}) = (120, 32)$ for A* and B*, but the runs behind Table 2 used $(32, 32)$, as `make c1` does by default. `HWT=120` runs with $h = 120$; see Appendix F of the full version.

(E2): *BFV pt-ct: Table 3* (C2) [3 compute-hours]

Execution: `make c2 THREADS=8`.

Results: Each of the five ring dimensions, $\log_2 N = 12, \ldots, 16$, must end with `SK Coeff Match Ratio : 1.000000` and `Final L2 Error (e) : 0`; the report tabulates them as Table 3, which is the claim. The eight columns describing the attack are fixed by the parameters and reproduce exactly; the two timing columns will not, ours running 1.06 to 2.2 times slower than the paper's native macOS build. Figure 3 is drawn from the run's own log as `results/figures/fig3-01.pdf`, the noise climbing past the certified bound.

Remarks: Some keys stop the recovery at the first index, as the noise they amplify runs past the $\log_2 t$-bit window the decrypted result has to land in. Over 100 draws at each ring dimension we saw this 13 times at $N = 2^{12}$ and 29 at 2^{16}, and never at the other three; the target then draws again, and the report says how many draws each dimension took.

`make c2` runs our tuning code, which also reads the noise with the secret key to pick and check its parameters. `make c2-oracle` [50 compute-minutes] uses the secret key only for key generation, the decryption oracle and grading.

(E3): *BFV ct-ct: Figures 4 and 5* (C3) [10 compute-minutes]

Execution: `make c3 THREADS=8`.

Results: `results/c3-ctct.txt` lists, per squaring, the bound and the two measured norms that Figure 4 plots. Then `results/figures/fig5-01.pdf`, drawn from the run, should look like Figure 5: both distributions centred at zero, told apart by their spread.

Remarks: About one run in eleven comes out flat, on a key the fixed gain does not suit. If the plot is flat, run it again. The IQR ratio of Section 5.2 depends on the key: on some, the IQR of $\mathcal{D}_0$ is 0 and the ratio undefined, while the two distributions remain apart.

A.5 Notes on Reusability

The attacks are parameterised on the command line rather than recompiled. In particular, `indcpad_ctct` takes a $\delta_{\text{ms}}/\sqrt{N}$ argument that lets one binary serve both the bound OpenFHE certified before its revision and the one it certifies now, so a future bound is a matter of changing a number; `ctct_multikey` likewise switches between a fixed and a rescaled gain by a flag.

The `Makefile` exposes `THREADS`, `TRIALS`, `FAILS` and `HWT`, and the Section 3 analysis takes `-m` for the parameter set. Applying the attacks elsewhere is a matter of the three knobs the paper describes: the gain C, which places the noise in the readout window; M, which buys signal-to-noise as $\sqrt{M}$; and the number of squarings.

A.6 Version

Based on the LaTeX template for Artifact Evaluation V20220926.